

DIAGONAL LOADING AND LOW-RANK STRATA AT THE FIRST OPEN ORDER OF LIEB'S PERMANENTAL-DOMINANCE CONJECTURE

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ABSTRACT. For an irreducible character χ^λ of $\mathfrak{S}_n$, Lieb's permanental-dominance conjecture asks whether the normalized immanant of every positive semidefinite Hermitian matrix is at most its permanent. Existing shape theorems leave a unique partition at the first unsettled order $n = 14$, namely $\lambda = (4, 4, 3, 3)$. We prove that its immanant vanishes on matrices of rank at most three. For Gram matrices whose vectors lie on at most four projective lines, we obtain the stronger unnormalized inequality $\text{Imm}_\lambda(A) \leq \text{per}(A)$. Our main structural result is a diagonal-loading principle: provided permanental dominance is known in all smaller orders, every verified matrix of order n remains verified after addition of an arbitrary nonnegative diagonal matrix. Consequently, the order-fourteen conjecture holds for $A = C + D$ whenever $\text{rank } C \leq 3$ and $D \geq 0$ is diagonal, and also when C is a four-line Gram matrix. We also show that the diagonal equality stratum is transversely stable, with quadratic coefficient $-92/91$ for the normalized immanant ratio. An exact-arithmetic search framework for the remaining strata is described; it found no counterexample, and is not used as evidence for the stated theorems.

1. INTRODUCTION

For a partition $\lambda \vdash n$, let χ^λ be the corresponding irreducible character of $\mathfrak{S}_n$, let $f^\lambda = \chi^\lambda(e)$, and define

$$\text{Imm}_\lambda(A) = \sum_{\sigma \in \mathfrak{S}_n} \chi^\lambda(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}. \quad (1)$$

Lieb's permanental-dominance conjecture asserts that

$$\text{Imm}_\lambda(A) \leq f^\lambda \text{per}(A) \quad (A \succeq 0). \quad (2)$$

The conjecture originates in Lieb's 1966 paper [3]; its history and its distinction from stronger, now disproved, "permanent-on-top" statements are surveyed by Wanless [7]. It remains open despite being known for all immanants through order 13 [4, 7].

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Pate proved (2) for broad families of shapes. In particular, his 1998 theorem covers shapes of the form $(p, q, r, 2^s, 1^t)$, and his 1999 refinement covers $(p, q^w, r, 2^s, 1^t)$ for $0 \leq w \leq 2$ [4, 5]. At order 14 these results leave only

$$\lambda = (4, 4, 3, 3), \quad f^\lambda = 12012. \quad (3)$$

Thus a search over all 135 partitions of 14 is unnecessary. More importantly, a rank-two or rank-three Gram search cannot see the remaining case at all: its target immanant is identically zero there.

The purpose of this paper is to identify and prove several positive strata at this frontier. The main results are as follows.

- (i) The target immanant vanishes in rank at most three (Proposition 3.1).
- (ii) On Gram matrices supported on at most four projective lines, the much stronger inequality $\text{Imm}_\lambda \leq \text{per}$ holds (Theorem 4.1).
- (iii) If all smaller orders satisfy permanental dominance, then validity at a base matrix C is preserved by every loading $C + D$ with $D \geq 0$ diagonal (Theorem 5.1).
- (iv) The diagonal equality stratum is a strict transverse local maximum of the normalized ratio (Proposition 6.1).

The loading result extends the rank-one-plus-diagonal class recorded by Rodtes [6, Proposition 2.6]; at order 14 it converts the rank-three vanishing result and the four-line theorem into families containing positive definite matrices.

2. THE UNIQUE FRONTIER SHAPE

We begin with a short reduction which fixes the target unambiguously.

Proposition 2.1. *Among the partitions of 14, the only shape not covered by the Pate families $(p, q, r, 2^s, 1^t)$ and $(p, q^2, r, 2^s, 1^t)$ is $(4, 4, 3, 3)$. Moreover, $f^{(4,4,3,3)} = 12012$.*

Proof. An uncovered partition must have at least four parts strictly exceeding 2. There cannot be five, since their sum would be at least 15. Write the four large parts as $a \geq b \geq c \geq d \geq 3$. The 1999 family applies when $b = c$. Otherwise $b \geq c + 1$, and hence

$$a + b + c + d \geq 4 + 4 + 3 + 3 = 14.$$

Equality is forced, leaving exactly $(4, 4, 3, 3)$ and no additional parts. The hook-length formula gives

$$f^{(4,4,3,3)} = \frac{14!}{\prod_{(i,j) \in (4,4,3,3)} h_{ij}} = 12012.$$

□

We henceforth fix $\lambda = (4, 4, 3, 3)$. Its conjugate partition is

$$\lambda' = (4, 4, 4, 2). \quad (4)$$

3. RANK OBSTRUCTION

The Gram interpretation of immanants connects their vanishing to symmetrized decomposable tensors. We use Gamas's criterion [1]: if $A = (\langle v_i, v_j \rangle)$, then $\text{Imm}_\lambda(A)$ is nonzero precisely when the vectors $v_1, \dots, v_n$ can be partitioned into linearly independent sets whose cardinalities are the column lengths of λ .

Proposition 3.1. *If $A \succeq 0$ and $\text{rank } A \leq 3$, then $\text{Imm}_{(4,4,3,3)}(A) = 0$. In particular, (2) holds on this rank stratum.*

Proof. Represent A as the Gram matrix of vectors in a space of dimension at most three. By (4), Gamas's criterion would require three linearly independent four-sets. No such four-set exists. Therefore the symmetrized tensor, and hence its squared norm $\text{Imm}_\lambda(A)$, is zero. The permanent of a positive semidefinite Hermitian matrix is nonnegative, so (2) follows. $\square$

Thus rank four is the first meaningful Gram-search stratum. The next theorem settles a structured but infinite part of it.

4. FOUR PROJECTIVE LINES

For a multi-index $\mu = (\mu_1, \dots, \mu_r)$ and an $r \times r$ matrix B , write B_μ for the matrix obtained by repeating row and column i exactly μ_i times.

Theorem 4.1. *Suppose that A is the Gram matrix of fourteen vectors contained in the union of at most four one-dimensional subspaces. Then*

$$\text{Imm}_{(4,4,3,3)}(A) \leq \text{per}(A). \quad (5)$$

In particular, this improves the conjectured bound on this class by the factor $f^{(4,4,3,3)} = 12012$.

Proof. Zero vectors make both sides vanish. Rescaling the individual vectors and changing their phases multiply both matrix functions by the same nonnegative factor, so it suffices to consider repeated copies of direction vectors $y_1, \dots, y_r$, where $r \leq 4$.

If the span has dimension at most three, Proposition 3.1 applies. Assume therefore that $r = 4$ and the four directions are independent. Every independent four-set contains one vector from each direction. Gamas's criterion and (4) show that the immanant can be nonzero only when the direction multiplicities are a permutation of $(4, 4, 3, 3)$. Relabel them so that $\mu = (4, 4, 3, 3)$, set

$$B = (\langle y_i, y_j \rangle)_{i,j=1}^4, \quad J = \{1, 2\}, \quad \mu! = 4! 4! 3! 3!.$$

The weight-space trace formula for immanants [2] yields

$$\text{Imm}_{(4,4,3,3)}(B_\mu) = \mu! \det(B)^3 \det(B[J, J]). \quad (6)$$

For completeness, the representation-theoretic reason is especially simple here:

$$S_{(4,4,3,3)}(\mathbb{C}^4) \cong \det^3 \otimes \bigwedge^2 \mathbb{C}^4,$$

and the weight $(4, 4, 3, 3)$ is the one-dimensional weight generated by $e_1 \wedge e_2$. The corresponding diagonal matrix coefficient is $\det(B)^3 \det(B[J, J])$; repetition contributes the factor $\mu!$.

Hadamard's determinant inequality gives

$$\det(B)^3 \det(B[J, J]) \leq b_{11}^4 b_{22}^4 b_{33}^3 b_{44}^3. \quad (7)$$

On the other hand, iterating Lieb's block permanent inequality [3] over the four repeated-direction diagonal blocks gives

$$\begin{aligned} \text{per}(B_\mu) &\geq \prod_{i=1}^4 \text{per}(b_{ii} \mathbf{1}_{\mu_i \times \mu_i}) \\ &= \mu! b_{11}^4 b_{22}^4 b_{33}^3 b_{44}^3. \end{aligned} \quad (8)$$

Combining (6)–(8) proves (5). $\square$

5. DIAGONAL-LOADING CLOSURE

The next result is the main structural observation. It is stated at arbitrary order because its proof is no harder than the specialization needed here.

For a partition $\alpha \vdash N$ and a positive semidefinite matrix C of order N , write $\text{PD}(\alpha, C)$ for the inequality

$$\text{Imm}_\alpha(C) \leq f^\alpha \text{per}(C).$$

Theorem 5.1 (Diagonal-loading principle). *Fix $\alpha \vdash N$ and assume that permanental dominance holds for every irreducible character of $\mathfrak{S}_k$ for all $k < N$. If $\text{PD}(\alpha, C)$ holds for a positive semidefinite matrix C of order N , then*

$$\text{PD}(\alpha, C + D)$$

holds for every nonnegative diagonal matrix D .

Proof. Write $D = \text{diag}(d_1, \dots, d_N)$ and $[N] = \{1, \dots, N\}$. Expanding the factors in (1) according to which diagonal entries are selected from D gives

$$\text{Imm}_\alpha(C + D) = \sum_{S \subseteq [N]} \left(\prod_{i \notin S} d_i \right) d_{\text{Res}_{\mathfrak{S}_S}^{\mathfrak{S}_N} \chi^\alpha}(C[S]). \quad (9)$$

Here $\mathfrak{S}_S$ denotes the permutations of S , extended by the identity on $[N] \setminus S$, and $C[S]$ is the principal submatrix on S .

Let $k = |S|$. The branching rule decomposes the restricted character as

$$\text{Res}_{\mathfrak{S}_S}^{\mathfrak{S}_N} \chi^\alpha = \sum_{\nu \vdash k} m_{S, \nu} \chi^\nu, \quad m_{S, \nu} \in \mathbb{Z}_{\geq 0}. \quad (10)$$

Restriction preserves degree, so

$$\sum_{\nu \vdash k} m_{S, \nu} f^\nu = f^\alpha. \quad (11)$$

For $S = \emptyset$, we use the conventions $\text{per}(C[\emptyset]) = 1$ and $d_{\text{Res } \chi^\alpha}(C[\emptyset]) = f^\alpha$. If $k < N$, the assumed lower-order inequalities and (10)–(11) imply

$$\begin{aligned} d_{\text{Res } \chi^\alpha}(C[S]) &= \sum_{\nu \vdash k} m_{S,\nu} \text{Imm}_\nu(C[S]) \\ &\leq \sum_{\nu \vdash k} m_{S,\nu} f^\nu \text{per}(C[S]) = f^\alpha \text{per}(C[S]). \end{aligned} \quad (12)$$

For $S = [N]$, the same bound is precisely $\text{PD}(\alpha, C)$.

The permanent has the parallel expansion

$$\text{per}(C + D) = \sum_{S \subseteq [N]} \left(\prod_{i \notin S} d_i \right) \text{per}(C[S]). \quad (13)$$

Every coefficient in (9) and (13) is nonnegative. Summing (12) over S proves the theorem. $\square$

Because all lower orders are known at $N = 14$, the theorem immediately produces two substantial classes at the unique frontier.

Corollary 5.2. *Let C, D be Hermitian matrices of order 14 with*

$$C \succeq 0, \quad \text{rank } C \leq 3, \quad D = \text{diag}(d_1, \dots, d_{14}) \succeq 0.$$

Then

$$\text{Imm}_{(4,4,3,3)}(C + D) \leq 12012 \text{per}(C + D).$$

If every $d_i > 0$, this gives a positive definite class.

Proof. Proposition 3.1 verifies the base matrix C , and Theorem 5.1 applies. $\square$

Corollary 5.3. *The same conclusion holds when C is a Gram matrix whose fourteen Gram vectors lie on at most four projective lines and $D \succeq 0$ is diagonal.*

Proof. Use Theorem 4.1 at the base point and then apply Theorem 5.1. $\square$

Remark 5.4. The proof of Theorem 5.1 is coefficientwise. It therefore does not require strict positivity of D , nonsingularity of C , or any limiting argument. It also shows that once the full conjecture is known through order $N - 1$, the set of verified order- N matrices is upward closed along every nonnegative diagonal ray.

6. LOCAL STABILITY OF THE DIAGONAL EQUALITY STRATUM

Diagonal matrices are equality cases in (2). The next expansion shows that a counterexample cannot bifurcate transversely from this stratum.

Proposition 6.1. *Let*

$$R(A) = \frac{\text{Imm}_{(4,4,3,3)}(A)}{12012 \text{per}(A)}.$$

If H is Hermitian with zero diagonal, then, as $H \rightarrow 0$,

$$R(I + H) = 1 - \frac{92}{91} \sum_{i < j} |h_{ij}|^2 + O(\|H\|^3). \quad (14)$$

Consequently, the identity is a strict local maximizer of R on the correlation-matrix slice. By positive diagonal congruence, every positive diagonal matrix has the same transverse local stability.

Proof. The identity permutation supplies the constant term. Since $h_{ii} = 0$, the only degree-two contributions come from transpositions, and therefore

$$\begin{aligned} \text{per}(I + H) &= 1 + \sum_{i < j} |h_{ij}|^2 + O(\|H\|^3), \\ \frac{\text{Imm}_\lambda(I + H)}{f^\lambda} &= 1 + \frac{\chi^\lambda(2, 1^{12})}{f^\lambda} \sum_{i < j} |h_{ij}|^2 + O(\|H\|^3). \end{aligned}$$

The normalized character value at a transposition is the normalized content sum:

$$\begin{aligned} \frac{\chi^\lambda(2, 1^{12})}{f^\lambda} &= \frac{\sum_i \binom{\lambda_i}{2} - \sum_j \binom{\lambda'_j}{2}}{\binom{14}{2}} \\ &= \frac{18 - 19}{91} = -\frac{1}{91}. \end{aligned}$$

Dividing the two expansions yields (14). The quadratic term is negative definite on the zero-diagonal Hermitian space, proving strict local maximality. Finally, if E is invertible diagonal, then both immanant and permanent are multiplied by $|\det E|^2$ under $A \mapsto EAE^*$. $\square$

7. EXACT COMPUTATIONAL ATTACK ON THE REMAINING STRATA

This section records the computational boundary of the work. None of its negative search results is used in the proofs above.

For an integer Gram factor $X \in \mathbb{Z}^{r \times 14}$, the matrix $A = X^*X$ is positive semidefinite and the counterexample condition is the strict integer inequality

$$\text{Imm}_{(4,4,3,3)}(A) > 12012 \text{ per}(A). \quad (15)$$

The permanent is evaluated by Ryser's formula. For the immanant, cycle weights for every vertex subset are computed by a Held–Karp path dynamic program. A second subset dynamic program combines those cycles with Murnaghan–Nakayama rim-hook transitions. Retained integer candidates are checked by a second implementation which first aggregates the entire weighted cycle-type distribution and applies the character only at the end. Thus an eventual certificate for (15) consists only of X and two exact integers; floating point is not part of the verification boundary.

The recorded search used unit-column real Gram factors and CMA–ES in hyperspherical coordinates. Table 1 lists representative best ratios. The rank-four entry was subsequently rounded and recomputed by both exact engines;

it remained far below 1. The values in the table are search diagnostics, not upper bounds.

TABLE 1. Representative numerical maxima of $R(A)$ found by the recorded searches. No row is a proof of global optimality.

Rank	Start/search family	Best recorded $R(A)$
4	clustered and continued CMA-ES	0.0004385601
5	random hyperspherical CMA-ES	0.0027358237
8	random hyperspherical CMA-ES	0.0474677393
11	random hyperspherical CMA-ES	0.1019042301
13	perturbed simplex	0.6065601015
14	random hyperspherical CMA-ES	0.1358999669
14	perturbed identity	0.8966876216

The low-rank theorem explains why ranks two and three never appear in this table. Proposition 6.1 also explains why a local optimizer near the identity returns toward, rather than through, the equality stratum.

8. CONCLUDING REMARKS

The original permanental-dominance conjecture remains open. At its first unsettled immanant order, however, the search space can be reduced sharply: only $(4, 4, 3, 3)$ remains, rank four is the first nonzero Gram stratum, four projective lines are safe, all nonnegative diagonal loadings of safe base points are safe, and the diagonal equality set is transversely stable.

Two concrete questions remain. First, does the stronger comparison

$$\text{Imm}_{(4,4,3,3)}(A) \leq \text{Imm}_{(4,4,4,2)}(A)$$

hold on rank-four Gram matrices? The two partitions are conjugate and have the same degree, and the comparison is consistent with all recorded tests, but no proof is claimed here. Second, can every correlation matrix in a substantial neighborhood beyond the local regime be decomposed into one of the verified bases plus a nonnegative diagonal loading? Either direction would enlarge the positive region without relying on exhaustive search.

COMPUTATIONAL DISCLOSURE

OpenAI Codex assisted with symbolic checks, implementation of the exact evaluators, numerical search, and language editing. The mathematical claims in the paper are supported by the displayed proofs; the computational experiments are explicitly separated from those proofs. All retained exact calculations are reproducible from the accompanying source files.

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